**LAYOUT - top view, front of the car to the RIGHT, driver side UP (R140 / R153). Not to scale; lengths in mm (estimates until measured at the car).**

Each segment: length | wires in it | DR-25 heat-shrink size for its bundle (est. from the wire ODs). Lengths per wire (path + 100 mm at each end) are in the pinout table.

^ DRIVER SIDE (LH) at the top / PASSENGER SIDE (RH) at the bottom v

## SEGMENTS

| FROM | TO | MM | WIRES | BUNDLE | DR-25 |
|---|---|---|---|---|---|
| ECU | SP | 110 | 114 | 30.5 | DR-25-1-1/2-0 |
| SP | BP_AC | 110 | 115 | 27.2 | DR-25-1-1/2-0 |
| BP_AC | BP_GB | 110 | 113 | 27.1 | DR-25-1-1/2-0 |
| BP_GB | BH | 100 | 64 | 18.3 | DR-25-1-0 |
| BP_GB | GB | 150 | 63 | 22.3 | DR-25-1-1/2-0 |
| GB | RHKICK | 250 | 14 | 10.6 | DR-25-1/2-0 |
| RHKICK | RHSILL | 1600 | 8 | 6.1 | DR-25-3/8-0 |
| RHSILL | FUEL | 400 | 3 | 5.3 | DR-25-1/4-0 |
| BP_AC | ACAMP | 400 | 2 | 2.9 | DR-25-3/16-0 |
| BP_AC | EVAP | 450 | 2 | 2.7 | DR-25-3/16-0 |
| ECU | BP_DASH | 30 | 61 | 27.9 | DR-25-1-1/2-0 |
| BP_DASH | DASH | 300 | 20 | 23.6 | DR-25-1-1/2-0 |
| BP_DASH | BP_COL | 220 | 48 | 25.1 | DR-25-1-1/2-0 |
| BP_COL | PEDAL | 640 | 11 | 15.8 | DR-25-3/4-0 |
| BP_COL | JB1 | 650 | 37 | 20.9 | DR-25-1-0 |
| BP_DASH | ALARM | 150 | 12 | 6.5 | DR-25-3/8-0 |
| ECU | CSB3 | 370 | 16 | 10.7 | DR-25-1/2-0 |
| ECU | CANJ | 50 | 17 | 16.6 | DR-25-1-0 |
| BH_ENG | BP_ENG_A1 | 100 | 41 | 24.5 | DR-25-1-1/2-0 |
| BP_ENG_A1 | CRANK | 1410 | 3 | 7.9 | DR-25-3/8-0 |
| BP_ENG_A1 | MRS | 1360 | 7 | 7.8 | DR-25-3/8-0 |
| BP_ENG_A1 | BP_ENG_A2 | 200 | 39 | 24.0 | DR-25-1-1/2-0 |
| BP_ENG_A2 | MAP | 130 | 3 | 3.3 | DR-25-3/16-0 |
| BP_ENG_A2 | IAT | 400 | 2 | 2.7 | DR-25-3/16-0 |
| BP_ENG_A2 | ETB | 430 | 8 | 15.3 | DR-25-3/4-0 |
| BP_ENG_A2 | FLEX | 230 | 4 | 10.8 | DR-25-1/2-0 |
| BP_ENG_A2 | CAM | 850 | 4 | 8.1 | DR-25-3/8-0 |
| BP_ENG_A2 | INJ4 | 160 | 8 | 6.3 | DR-25-3/8-0 |
| INJ4 | INJ3 | 180 | 6 | 5.5 | DR-25-1/4-0 |
| INJ3 | INJ2 | 380 | 4 | 4.5 | DR-25-1/4-0 |
| INJ2 | INJ1 | 170 | 2 | 3.2 | DR-25-3/16-0 |
| BP_ENG_A2 | COIL4 | 100 | 12 | 7.7 | DR-25-3/8-0 |
| COIL4 | COIL3 | 160 | 9 | 6.7 | DR-25-3/8-0 |
| COIL3 | COIL2 | 180 | 6 | 5.5 | DR-25-1/4-0 |
| COIL2 | COIL1 | 180 | 3 | 3.9 | DR-25-3/16-0 |
| BP_ENG_A2 | ENG_BLOCK | 300 | 5 | 8.3 | DR-25-1/2-0 |
| BP_ENG_A2 | BP_ENG_A3 | 450 | 8 | 11.4 | DR-25-3/4-0 |
| BP_ENG_A3 | TURBO | 150 | 4 | 10.8 | DR-25-1/2-0 |
| BP_ENG_A3 | BOOST | 280 | 2 | 2.7 | DR-25-3/16-0 |
| BP_ENG_A3 | ECT | 700 | 2 | 2.7 | DR-25-3/16-0 |
| BH_ENG | BP_ENG_B0 | 150 | 7 | 11.0 | DR-25-1/2-0 |
| BP_ENG_B0 | LAMJ | 90 | 5 | 10.6 | DR-25-1/2-0 |
| LAMJ | LAMBDA | 150 | 4 | 3.8 | DR-25-3/16-0 |
| BP_ENG_B0 | OILT | 960 | 2 | 2.7 | DR-25-3/16-0 |
| BH_ENG | BP_ENG_B1 | 200 | 18 | 14.9 | DR-25-3/4-0 |
| BP_ENG_B1 | KNOCK | 1260 | 3 | 9.4 | DR-25-1/2-0 |
| BP_ENG_B1 | BP_ENG_B2 | 200 | 15 | 11.7 | DR-25-3/4-0 |
| BP_ENG_B2 | FUELP | 30 | 3 | 3.3 | DR-25-3/16-0 |
| BP_ENG_B2 | OILP | 580 | 3 | 3.3 | DR-25-3/16-0 |
| BP_ENG_B2 | CLNTP | 1020 | 3 | 3.3 | DR-25-3/16-0 |
| BP_ENG_B2 | BP_ENG_B3 | 300 | 8 | 11.7 | DR-25-3/4-0 |
| BP_ENG_B3 | CHIAT | 360 | 2 | 2.7 | DR-25-3/16-0 |
| BP_ENG_B3 | VSS | 680 | 4 | 10.8 | DR-25-1/2-0 |
| BP_ENG_B3 | REVSW | 880 | 2 | 3.6 | DR-25-3/16-0 |
| GB | FW_STR | 400 | 5 | 6.0 | DR-25-3/8-0 |
| FW_STR | STARTER | 2470 | 1 | 4.6 | DR-25-1/4-0 |
| CANJ | CLUSTER | 400 | 11 | 11.2 | DR-25-3/4-0 |
| CANJ | PI | 250 | 6 | 9.0 | DR-25-1/2-0 |
| BP_COL | COLUMN | 400 | 3 | 3.3 | DR-25-3/16-0 |
| ALARM | VALET | 250 | 2 | 2.7 | DR-25-3/16-0 |
| GB | RFG | 300 | 27 | 16.1 | DR-25-3/4-0 |
| RFG | C_FEND | 600 | 24 | 16.0 | DR-25-3/4-0 |
| C_FEND | ACC | 250 | 2 | 3.4 | DR-25-3/16-0 |
| C_FEND | MRS | 400 | 4 | 6.9 | DR-25-3/8-0 |
| C_FEND | C_RHC | 500 | 20 | 14.3 | DR-25-3/4-0 |
| C_RHC | FANC | 300 | 4 | 12.2 | DR-25-3/4-0 |
| FANC | FANR | 300 | 2 | 8.6 | DR-25-1/2-0 |
| C_RHC | C_FRONT | 700 | 19 | 11.5 | DR-25-3/4-0 |
| C_FRONT | C_LHC | 700 | 11 | 8.8 | DR-25-1/2-0 |
| RFG | SIREN | 500 | 2 | 2.7 | DR-25-3/16-0 |
| RFG | ALT | 400 | 3 | 3.7 | DR-25-3/16-0 |
| RHKICK | RFP | 300 | 2 | 7.6 | DR-25-3/8-0 |
| JB1 | LFP | 300 | 6 | 9.6 | DR-25-1/2-0 |
| LFP | WIPER | 300 | 4 | 5.8 | DR-25-3/8-0 |
| JB1 | LHSILL | 1800 | 10 | 12.2 | DR-25-3/4-0 |
| LHSILL | RLJ | 800 | 6 | 5.7 | DR-25-3/8-0 |
| RLJ | R11 | 300 | 6 | 6.1 | DR-25-3/8-0 |
| RLJ | RPANEL | 600 | 12 | 8.5 | DR-25-1/2-0 |
| RPANEL | R12 | 600 | 6 | 6.1 | DR-25-3/8-0 |
| RPANEL | HATCH | 400 | 2 | 3.6 | DR-25-3/16-0 |
| LHSILL | WSRL | 400 | 2 | 7.6 | DR-25-3/8-0 |
| LHSILL | WSRR | 1300 | 2 | 7.6 | DR-25-3/8-0 |
| FUEL | TANK | 300 | 3 | 5.3 | DR-25-1/4-0 |
| RHSILL | BATT | 600 | 7 | 7.1 | DR-25-3/8-0 |
| RL_ENG | ENG_BLOCK | 690 | 3 | 3.3 | DR-25-3/16-0 |
| RL_ENG | STARTER_B | 1500 | 2 | 2.7 | DR-25-3/16-0 |
| ALT_B | RL_ENG | 300 | 1 | 1.9 | DR-25-3/16-0 |
| STARTER_B | JUMP | 210 | 1 | 1.9 | DR-25-3/16-0 |

## NODES

ECU ECU plugs A / B (centre dash, factory ECM spot)
SP ECU splice point (bussed DT plugs + soldered splices)
BP_AC A/C branch
BP_GB glove-box branch
BH DRB102 bulkhead, cabin half (RH firewall)
GB Glove box: POWER + LOGIC boxes (every cabin fuse and relay), NEG bar, SW12 bus, blower lead
RHKICK RH kick panel (old R/B No.4 spot): ST1 / heater taps, ground IG
RHSILL RH sill, rear end (the run turns across the trunk)
FUEL Trunk LH: IX-FUEL inline to the fuel sender
ACAMP A/C amplifier (A18) taps
EVAP Evaporator case: thermistor A24 leads
BP_DASH dash branch
DASH Mid-dash above the ECU: front + rear WS inlines, A/C switch lead
BP_COL pedal / column branch
PEDAL Accelerator pedal (driver footwell)
JB1 J/B No.1 + cowl plug IE1, LH kick panel: IG2 / tail / turn taps, door switch lead, ground ID (Routes 2, 7, 15)
ALARM Anti-theft unit (centre dash, left of the ECU)
CSB3 Centre console: CSB3 under the centre cluster, its switch inputs and opto board; alarm status LED (Routes 12, 14)
CANJ CAN trunk at the ECU: CAN H / L joins, ECU 120 ohm end (Route 8)
BH_ENG DRB102 bulkhead, engine half (RH firewall)
BP_ENG_A1 A trunk branch point FP1 (just off the bulkhead, passenger side; Route 3 point 3)
BP_ENG_A2 A trunk branch point FP2 (top of the head, centre; Route 3 point 4)
BP_ENG_A3 A trunk branch point FP3 (driver side of the head; Route 3 point 5)
CRANK Crank VR sensor (Dan's 'crank below' spot; Route 3 point 6)
MRS MR-S EHPS pump plugs B / C (pump at the old ABS block spot; Route 3 point 7)
MAP MAP sensor (Route 3 point 9)
IAT IAT sensor, intake manifold (Route 3 point 8)
ETB Bosch 74.5 mm ETB (Route 3 point 10)
FLEX Flex fuel sensor (driver side, top; Route 3 point 12)
CAM Cam Hall sensor (centre of the head, behind the COP row; Route 3 point 11)
INJ4 Injector 4 (injector row enters at cyl 4; Route 3 point 13)
INJ3 Injector 3
INJ2 Injector 2
INJ1 Injector 1
COIL4 COP 4 (coil row enters at cyl 4; Route 3 point 14)
COIL3 COP 3
COIL2 COP 2
COIL1 COP 1
ENG_BLOCK Engine block ground stud (RADLOK engine ground H6; Route 1 point 5)
TURBO Turbo speed sensor (Route 3 point 17)
BOOST Boost control solenoid (Route 3 point 16)
ECT ECT sensor (Route 3 point 15)
BP_ENG_B0 B trunk branch point FP0 (passenger leg; Route 4 point 3)
LAMJ CAN-Lambda joins: CAN H / L, +12 V and ground splices (240 mm from the bulkhead)
LAMBDA Link CAN-Lambda controller (wideband O2; Route 4 point 8)
OILT Oil temp sender (Route 4 point 7)
BP_ENG_B1 B trunk branch point FP1 (rear of block, centre; Route 4 point 4)
KNOCK Knock sensor 1 (factory K1 leader; Route 4 point 9)
BP_ENG_B2 B trunk branch point FP2 (rear of block, driver side; Route 4 point 5)
FUELP Fuel pressure sensor (Route 4 point 10)
OILP Oil pressure sensor (Route 4 point 12)
CLNTP Coolant pressure sensor (rear of block, driver side; Route 4 point 11)
BP_ENG_B3 B trunk branch point FP3 (driver side, low; Route 4 point 6)
CHIAT Charge-pipe IAT sensor (Route 4 point 13)
VSS Gearbox VSS (spot estimated in Route 4, point 14)
REVSW Back-up (reverse) switch B1 on the transaxle - spot TBD (not on Route 4)
FW_STR RH firewall: RADLOK pair, cabin side, and the starter S wire pass - spot TBD (Route 1 point 1)
STARTER Starter solenoid S (50) post (starter: lower-middle of the engine; Route 1 point 2)
CLUSTER Centre cluster (old meter spot): LED board on the meter plugs C11 / C12, cluster CAN board + buck, J/B No....
PI RealDash Pi + USB-CAN adapter, centre dash, and the dash 12 V split (Route 8)
COLUMN Steering column: cruise stalk C14 flying leads (Route 12, spot approximate)
VALET Valet / program switch lead, under the carpet to the driver-side carpet centre, service coil (Route 14)
RFG Existing factory RH fender grommet, RH front inner fender corner: harness C, alarm siren + hood pin, alternator IG ...
RFP RH fender pass-through by the strut tower: FR wheel-speed plug (Route 9 rev 2)
LFP LH fender pass-through by the strut tower, driver kick: FL wheel-speed plug; wiper feed out to the cowl (Routes 2, 9)
WIPER Front wiper motor F9, under the cowl, driver side (Route 2 point 18)
LHSILL LH sill, rear end: under the rear seat (rear wheel-speed legs meet; rear-light run turns to the quarter panel; Rout...
RLJ LH quarter panel centre (OEM joint BP1): rear-light splices (Route 15)
R11 LH rear lamp R11 (socket pigtails)
RPANEL Rear panel centre: licence lamp L1, ground point BL, trunk pin switch (Routes 14, 15)
HATCH High-mount stop light H6 (hatch spoiler)
R12 RH rear lamp R12 (socket pigtails)
WSRL RL ABS sensor plug (rear left; Route 10)
WSRR RR ABS sensor plug (rear right; its cable crosses under the rear seat to the LH sill, Route 10)
TANK Fuel tank: fuel pump and the level sender (DT 2-way), trunk LH ground stud (Route 11 rev 2)
BATT Trunk RH rear corner: battery, kill switch, battery relay block (FP 40 A fuse + fuel pump relay K-FP) (Route 1 point...
C_FEND Harness C: down the RH inner fender (A/C branch, EPS pump) (Route 2 rev 12)
ACC A/C compressor A4 clutch lead + dual pressure switch A5 (Route 2 point 7)
C_RHC RH front corner: headlamp H2, turn / clearance F8, retract motor R3, washer W1, hood pin, ground EA (Route 2 ...
FANC Condenser fan (own connector; first stop on the fan branch, Route 2 point 9)
FANR Radiator fan (own connector; end of the fan branch, Route 2 point 10)
C_FRONT Core support, front centre: horn split, both horns (DT 2-way each), headlamp / retract / clearance return joins ...
C_LHC LH front corner: headlamp H1, turn / clearance F7, retract motor R2, ground EB (Route 2 point 16)
SIREN Alarm siren: centre of the firewall behind the engine, leads through the RH fender grommet (Route 14)
ALT Alternator G2 round 3-pin plug (IG / L / S, through the RH fender grommet) - rear RH of the engine under the bulkhe...
RL_ENG RADLOK + / - pair, engine side (RH firewall) (Route 1 point 1)
ALT_B Alternator B+ stud, rear RH of the engine (Route 1 point 3)
STARTER_B Starter B+ stud (H4, H5, H8 lugs) (Route 1 point 2)
JUMP Engine-bay jump post (Route 1 point 6, proposed spot)

The ECU splice point (SP) is ONE place by the ECU plugs (Dan 2026-10-19): +5 V on BUS-5V + BUS-5V-2, Gnd Out A24 on BUS-GNDOUT-A, B22 on BUS-GNDOUT-B (bussed DEUTSCH DT, DECISIONS R169 parts), +8 V, switched 12 V, the A5 feed and the ECU power ground soldered there. The ECU ground runs alone to the POWER box NEG bar (R216).

**REVISIONS**

| REV | DATE | BY | DESCRIPTION |
|---|---|---|---|
| A | 2026-10-20 | DG | First issue: every harness of the master in one drawing set (replaces the ST185-AB-cabin pilot, R... |

## ST185-harness - every wiring harness of the car, one drawing set

Layout - top view, segment lengths, wire counts, DR-25 sizes

| DRAWING NO. | REV | SHEET | SIZE | DRAWN | DATE |
|---|---|---|---|---|---|
| ST185-harness-LAY | A | 1 OF 1 | 1647 x 950 mm | DG | 2026-10-20 |

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale